

33S3 — Cross-Scale Performance, Recoverability, and Hysteretic Readdressing

Professional reading edition of the evolving public MKUFT canon

Mark Charles McLaughlin

ORCID: [0009-0005-7736-1511](https://orcid.org/0009-0005-7736-1511)

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1. Purpose

Module 33S2 separates functional identity, relational closure, and property-specific law descent. A further ambiguity appears whenever a lower-scale process becomes locally more successful while the enclosing higher-order organisation is changing:

Does improvement of a lower-address performance variable imply improvement of the enclosing whole?

No such implication is generally licensed.

A cell lineage may proliferate faster while tissue-level organisation worsens. A constituent process may gain local competitive advantage while collective-level fitness falls. Conversely, lower-level performance can also support or improve higher-level closure when the objectives and constraints are aligned. Therefore the sign and scale of improvement must be typed rather than inherited.

A second ambiguity concerns recovery:

If a higher-order organisation is degraded, does a route back necessarily remain available, and is that route the reverse of the route out?

Again, neither claim is generally licensed. Recoverability depends on the current state, target relation, allowed intervention class, environment, time horizon, and path/history variables. Some transitions are reversible, some are hysteretic, some require a different restoration route, and some leave the declared recoverable set altogether.

This module therefore adds two operational distinctions:

1. **cross-scale performance alignment** — whether local change supports, opposes, or is decoupled from enclosing-scale closure and law descent;
2. **conditional recoverability** — whether a damaged or displaced addressed state can lawfully reach a declared closure target under specified controls and bounds.

Compressed rule:

Local gain is not inherited as whole-level gain. Degradation is not inherited as inevitable collapse. Recovery is earned by reachability under declared conditions.

2. Typed local performance versus whole-level closure

Let

$$\pi_R : X \rightarrow Y$$

be the relational address map of Module 33S2, with lower state $x \in X$ and higher relational state $y = \pi_R(x)$.

Let

$$J_\ell : X \rightarrow \mathbb{R}$$

be a declared lower-address performance variable. Depending on the system, J_ℓ may represent growth rate, replication rate, throughput, local competitive success, immediate reward, resource capture, or another operational quantity. The notation does **not** imply conscious optimisation.

Retain separately:

$$M_C : Y \rightarrow \mathbb{R}$$

as a declared margin to loss of the constitutive closure relation, and

$$M_D(q) : Y \rightarrow \mathbb{R}$$

as the margin to loss of higher-address sufficiency for target property q .

For a matched comparison interval, record the change tuple

$$\Delta_{\ell \rightarrow h} = (\Delta J_\ell, \Delta M_C, \Delta M_D(q)).$$

The tuple is preferable to a single scalar because the three quantities may have different units and meanings.

Cross-scale alignment classes

For a declared interval and environment:

- **aligned:** local performance improves while the relevant whole-level margin is preserved or improves;
- **antagonistic:** local performance improves while the relevant whole-level margin decreases;
- **neutral/decoupled:** material local change occurs without a demonstrated change in the relevant whole-level margin;
- **mixed/property-relative:** alignment differs across closure and different target properties.

The important antagonistic pattern is therefore

$$\Delta J_\ell > 0, \quad \Delta M_C < 0$$

or, for a property-specific effect,

$$\Delta J_\ell > 0, \quad \Delta M_D(q) < 0.$$

These are possible cross-scale relations, not universal laws.

A process can become locally more successful while the enclosing object becomes less robust. The opposite can also occur. The sign must be measured at each address.

3. Why a universal “local gain causes collapse” rule fails

The following stronger rule is rejected:

$\Delta J_\ell > 0 \implies$ eventual higher-scale collapse.

It fails for several reasons:

1. local and higher-scale objectives can be aligned;
2. local gain can be irrelevant to the load-bearing higher relation;
3. compensating constraints can preserve or increase M_C ;
4. a locally costly constituent change can increase higher-level viability;
5. an apparent conflict can disappear after the lower or higher performance variable is correctly addressed;
6. the higher object may reconfigure rather than collapse;
7. a degraded state may remain recoverable.

The lawful statement is narrower:

Local performance and whole-level closure are distinct addressed variables whose coupling must be established rather than assumed.

4. Recoverability is a reachability claim

Let $\mathcal{C}_R \subseteq Y$ be a declared target set of higher states that satisfy the required closure relation R to a stated tolerance.

Let U be an allowed intervention/control class, E the allowed environment class, $\mathcal{D}$ the admissible state/path domain, and H a declared recovery horizon.

A lower-address state x_t is **recoverable to $\mathcal{C}_R$ over horizon H** when there exists an admissible control/intervention sequence whose resulting trajectory remains inside the declared admissible domain and reaches the target closure set:

$$\text{Rec}_H(x_t; \mathcal{C}_R, U, E) = 1$$

if and only if there exists an admissible route such that

$$\pi_R(x_{t+H}) \in \mathcal{C}_R.$$

Otherwise

$$\text{Rec}_H = 0$$

for that target, horizon, intervention class, and environment.

This is a reachability scaffold, not a claim to have invented viability or recoverability theory.

Recovery is target-relative

A system may recover:

- function without exact microstate;
- closure without the same constituent realization;
- one property but not another;
- the same relational class through a different trajectory;
- only under active intervention;
- only inside a bounded environmental regime.

Therefore “returned” must name the target equivalence class. Exact microscopic restoration must not be inferred from functional recovery.

5. Degradation, closure failure, and loss of recoverability are different events

Keep at least three boundaries separate:

1. **law-descent degradation:** $M_D(q)$ crosses the declared sufficiency threshold for property q ;
2. **closure failure:** the constitutive relation leaves the declared closure class or M_C crosses its failure threshold;
3. **recoverability loss:** the current state leaves the recoverable set for the declared target, intervention class, environment, and horizon.

No universal ordering is assumed.

For example, a system may satisfy

$$M_D(q) < 0, \quad M_C > 0, \quad \text{Rec}_H = 1,$$

meaning that a simple higher-level law for q has failed, the organised whole remains intact, and a declared recovery route remains available.

A different system may retain some recognizable organisation while already having

$$\text{Rec}_H = 0$$

for the chosen target and admissible controls.

Reduced predictability is not collapse; reduced closure is not automatically irreversibility; irreversibility must be tested against a declared recovery target and route class.

6. Hysteresis and asymmetric return paths

A restoration route need not be the time-reverse of the degradation route.

Let λ be a controlled parameter. If transition into one addressed regime occurs at threshold $\lambda_{\uparrow}$ while return occurs at a distinct threshold $\lambda_{\downarrow}$ under matched protocol, then

$$\lambda_{\uparrow} \neq \lambda_{\downarrow}$$

is a candidate hysteresis signature.

However, apparent hysteresis may reflect omitted state.

Before treating history dependence as irreducible, test whether augmenting the address with a bounded internal/history state

$$Y_t^+ = (Y_t, H_t^{(k)})$$

restores predictive closure. If a measurable hidden state explains the apparent path dependence, that state belongs in the address.

Thus:

Path dependence earns explicit history only after simpler state augmentation and ordinary physical memory variables have been tested.

Where hysteresis survives the declared state description, the lawful readdressing route is generally

```
state A
→ degradation path  $\gamma_{out}$ 
→ displaced state B
→ recovery path  $\gamma_{back}$ 
→ target relational class A'
```

with no requirement that

$$\gamma_{back} = \gamma_{out}^{-1}.$$

Nor is exact equality $A' = A$ required unless exact state restoration is the declared target.

7. Cross-domain stress tests

The purpose of these comparisons is not to claim one physical mechanism across domains. They test whether the distinctions above are needed independently.

7.1 Experimental transitions in individuality

Experimental work on multicellularity has demonstrated that collective-level fitness can become partly decoupled from constituent-cell performance and that within-collective conflict can undermine higher-level adaptation. In one experimental life-cycle system, higher lineage fitness improved while single-cell performance remained unchanged or declined under a collective-level selection regime. Snowflake-yeast work likewise shows that collective-level adaptation depends on how lower-level variation and conflict are organised.

This supports the need to type performance by scale. It does not establish MKUFT as the cause of those results.

Relevant primary literature:

- Hammerschmidt, K., Rose, C. J., Kerr, B. & Rainey, P. B. “Life cycles, fitness decoupling and the evolution of multicellularity.” *Nature* 515, 75–79 (2014). DOI: [10.1038/nature13884](https://doi.org/10.1038/nature13884).
- Ratcliff, W. C. et al. “Origins of multicellular evolvability in snowflake yeast.” *Nature Communications* 6, 6102 (2015). DOI: [10.1038/ncomms7102](https://doi.org/10.1038/ncomms7102).

7.2 Epithelial competition and tissue integrity

Epithelial systems provide a second hostile comparison because relative cellular competitive success and tissue-level integrity are not interchangeable. Normal epithelia can actively eliminate transformed cells, and the fate of emerging transformed populations depends on competition with surrounding tissue. In other contexts, competitive mechanisms can be altered or exploited in ways that no longer support tissue-level homeostasis.

The relevant structural lesson is limited:

cell-level competitive performance and tissue-level closure can align or conflict depending on the relational context.

Relevant primary literature:

- Kon, S. et al. “Cell competition with normal epithelial cells promotes apical extrusion of transformed cells through metabolic changes.” *Nature Cell Biology* 19, 530–541 (2017). DOI: [10.1038/ncb3509](https://doi.org/10.1038/ncb3509).
- Colom, B. et al. “Mutant clones in normal epithelium outcompete and eliminate emerging tumours.” *Nature* 598, 510–514 (2021). DOI: [10.1038/s41586-021-03965-7](https://doi.org/10.1038/s41586-021-03965-7).

These examples also block a simplistic mapping in which increased local competitive fitness is always harmful to the enclosing tissue. The sign is context-dependent, which is exactly why an explicit cross-scale alignment test is required.

7.3 Ecological recovery and hysteresis

Ecological systems provide a direct test of the claim that degradation must imply immediate or inevitable collapse. Perturbed systems can retain recovery capacity, recovery rates can change as a transition is approached, and alternative stable states can exhibit history-dependent transition paths.

Relevant primary literature:

- Rindi, L., Dal Bello, M., Dai, L., Gore, J. & Benedetti-Cecchi, L. “Direct observation of increasing recovery length before collapse of a marine benthic ecosystem.” *Nature Ecology & Evolution* 1, 0153 (2017). DOI: [10.1038/s41559-017-0153](https://doi.org/10.1038/s41559-017-0153).
- Zou, Y. et al. “Positive feedbacks and alternative stable states in forest leaf types.” *Nature Communications* 15, 4658 (2024). DOI: [10.1038/s41467-024-48676-5](https://doi.org/10.1038/s41467-024-48676-5).

These results do not imply that all systems possess tipping points or hysteresis. They demonstrate that resistance, recovery, alternative states, and history dependence are empirically separable phenomena and therefore should not be collapsed into a single “coherence” variable.

8. Hostile counterexample pass

The proposed refinement survives only in the narrow form above.

Counterexample A — aligned local and global improvement

If local performance increases and higher-level closure also improves, the framework must classify the episode as aligned rather than forcing a hidden cost.

Result: survives by rejecting a universal antagonism law.

Counterexample B — local loss supports the whole

A constituent can lose local performance while improving higher-level viability, as can occur when a collective-level life cycle suppresses lower-level conflict.

Result: survives because local and higher performance are separately addressed.

Counterexample C — apparent collapse precursor that recovers

A system can show degraded higher-level prediction or slowing recovery and nevertheless return to its target closure class.

Result: survives because $M_D(q)$, M_C , and recoverability are distinct.

Counterexample D — genuine irreversible transition

A state can leave the recoverable set for the declared target and controls.

Result: survives because recovery is conditional, not guaranteed.

Counterexample E — apparent hysteresis disappears after state augmentation

If adding a measurable internal state removes path dependence, the correct action is to readdress the system with the missing state rather than claim irreducible memory.

Result: survives by making history a burden, not a permission.

Counterexample F — same functional recovery, different microstate

A system can regain the same function through a different constituent realization.

Result: survives because recovery targets a declared relational/functional class unless exact token-state restoration is explicitly required.

9. Operational test sequence

For any claimed cross-scale degradation/recovery process:

1. identify the lower and higher addresses;
2. define the lower performance variable J_ℓ without assuming it is the whole-system objective;
3. define the load-bearing higher relation and an operational closure criterion;
4. define target property q and its law-descent criterion;
5. measure or estimate ΔJ_ℓ , ΔM_C , and $\Delta M_D(q)$ under matched conditions;
6. classify the cross-scale relation as aligned, antagonistic, neutral/decoupled, or mixed;
7. define the target recovery set $\mathcal{C}_R$;
8. declare admissible controls, environment, safety/domain constraints, and recovery horizon;
9. test recoverability prospectively rather than inferring it from apparent similarity to an earlier state;
10. test whether apparent history dependence disappears after bounded state augmentation;
11. if the old address no longer predicts the transition, readdress downward or multiscale under Module 33S2;
12. do not promote a domain-specific recovery mechanism into a universal law.

10. Falsification and reduction rules

This module must be reduced or treated as non-contributory where:

1. the lower performance variable and higher closure variable cannot be independently operationalised;
2. an apparent cross-scale conflict disappears under correct scale matching and no residual distinction remains;
3. recoverability is asserted without a declared target, intervention class, environment, horizon, or admissible route;
4. a successful return in one case is generalized into guaranteed recoverability;

5. decreased recovery rate is treated as proof of inevitable collapse;
6. an apparent hysteresis loop disappears when an omitted physical state variable is included, but irreducible history is still claimed;
7. a different microscopic realization is mislabeled as failed recovery despite meeting the declared functional target;
8. the same cross-domain vocabulary is used to claim an identical mechanism without mechanism-level evidence;
9. local performance is treated as inherently beneficial or inherently destructive at the enclosing scale.

11. Integration with the SIPO capstone

Module 33 constructs the addressed physical law object

$$\mathcal{L}_{P,t} = (\mathcal{D}_{P,t}, \mathcal{T}_{P,t}, \mathcal{W}_{P,t}).$$

Module 33S2 decides whether a higher relational address has earned law sufficiency for property q .

Module 33S3 adds a transition audit when locally improving dynamics may alter enclosing-scale closure or recoverability:

```
addressed state
→ local/higher performance variables separated
→ cross-scale alignment classified
→ closure and law-descent margins tracked separately
→ recovery target and admissible route class declared where relevant
→ propagate under the currently earned law address
→ test recovery / hysteresis / state augmentation
→ readdress if the active scale or sufficient state changes
```

This does not add a new fundamental force, coupling, or physical layer. It constrains how scale-relative performance and recovery claims are represented inside the existing addressed-law architecture.

12. Contribution boundary

This module does not claim invention of multilevel selection, fitness decoupling, resilience, recovery, viability kernels, reachability, hysteresis, alternative stable states, state augmentation, coarse-graining, or multiscale control.

The candidate MKUFT contribution is the operational conjunction:

within one addressed-law architecture, separate lower-scale performance from higher-scale closure and law-descent margins; permit aligned, antagonistic, neutral, and mixed cross-scale change; define recovery as conditional target-relative reachability rather than inevitable reversal; require apparent hysteresis to survive state-augmentation tests; and use the result to trigger bidirectional or multiscale readdressing without denying the reality of the higher-order object.

Historical priority for that exact conjunction is not asserted without broader review.

13. Compressed canonical rule

Precise form:

A locally improving process may support, oppose, or be irrelevant to the closure of the enclosing whole. Measure the signs at their own addresses. Loss of a property-specific higher law, loss of closure, and loss of recoverability are distinct events. A return route exists only when the current state can reach a declared target closure class under stated controls, environment, horizon, and admissibility constraints; the return path need not reverse the degradation path, and apparent history dependence must first survive state augmentation.

Mnemonic:

Local gain is not whole gain; recovery is a reachability claim.